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(54) **HANGER DEVICES**

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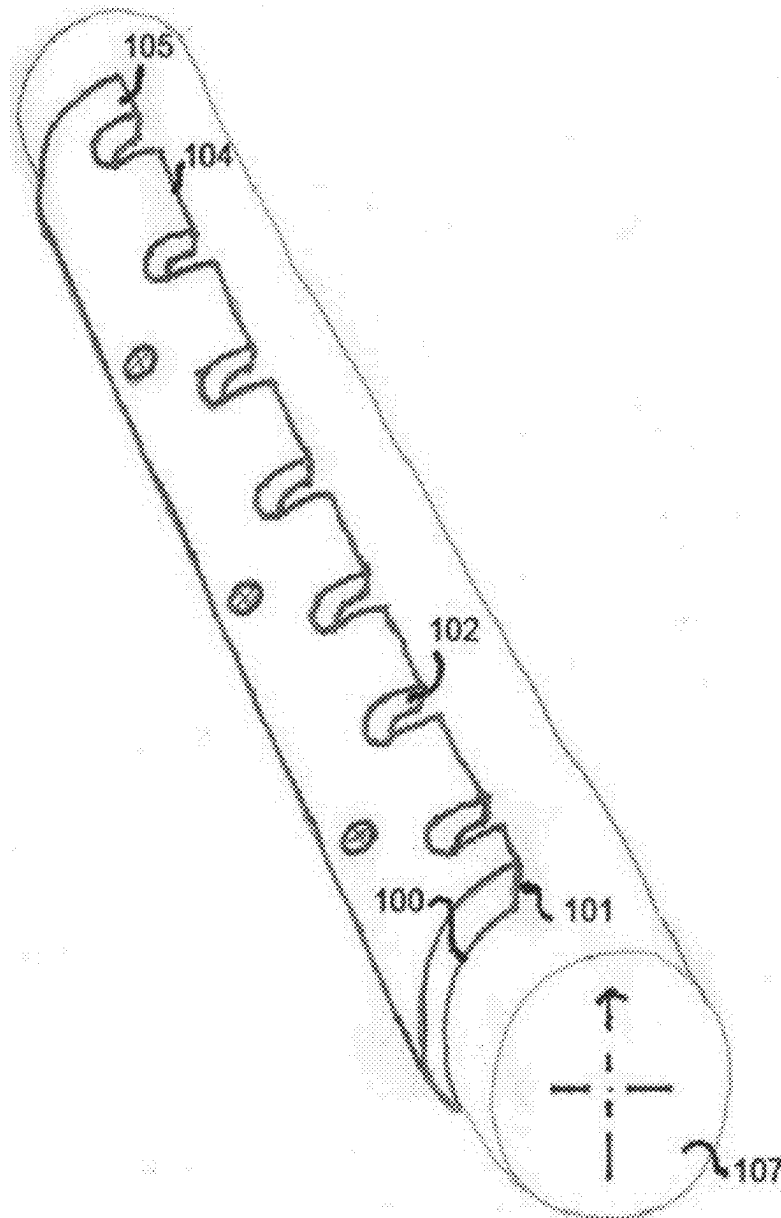
(57) **ABSTRACT**

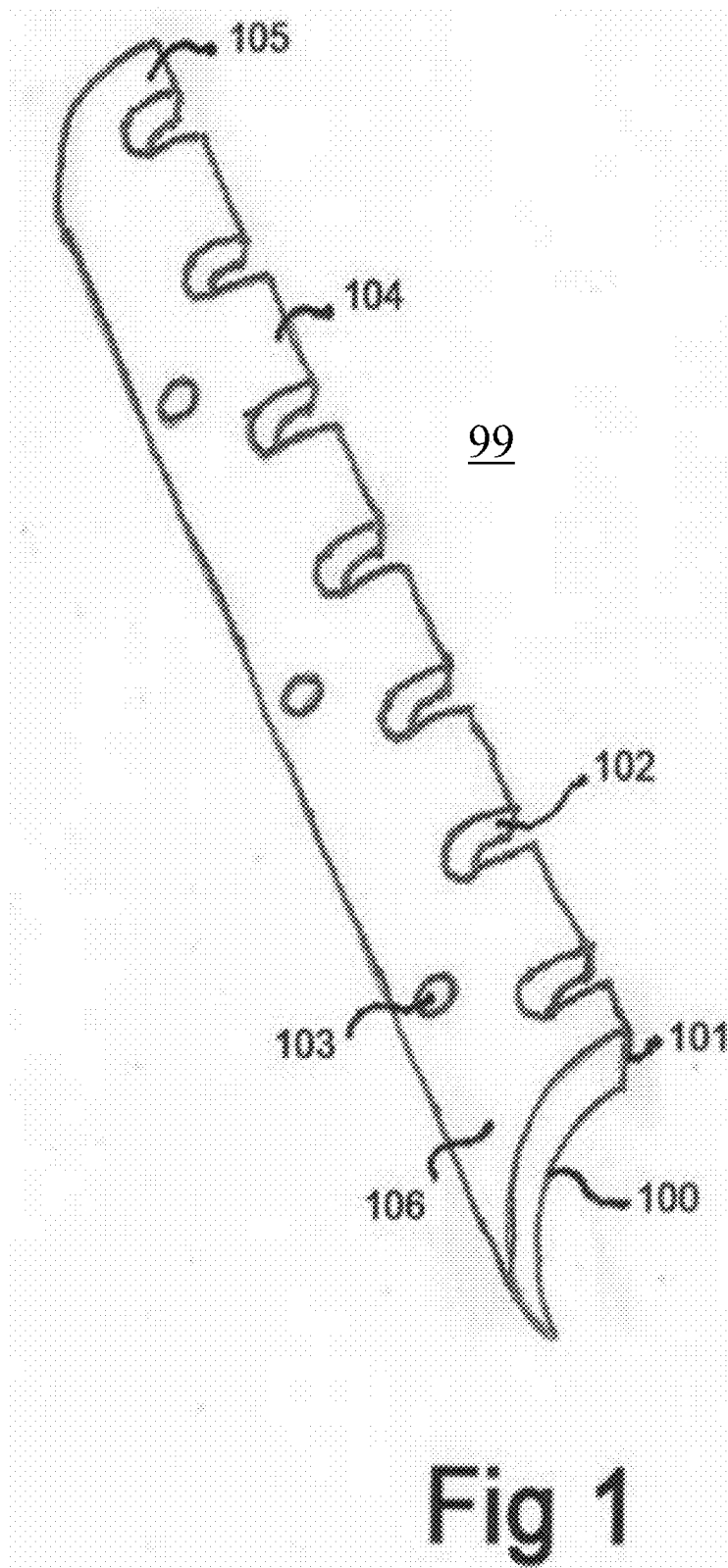
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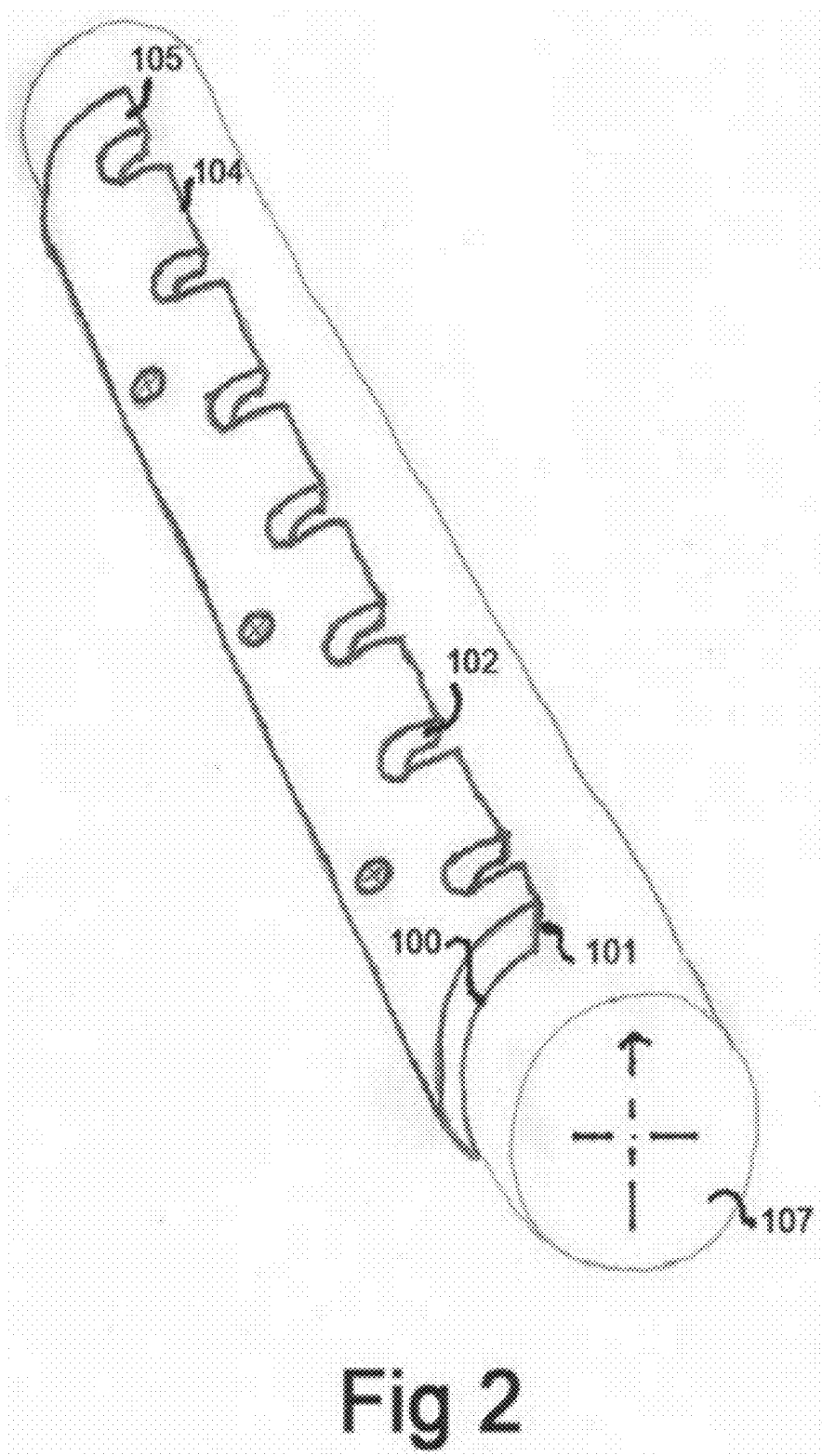
A device can include a body configured to attach to a hanging bar, and a plurality of hanger detents disposed in the body and configured to receive a portion of a hanger hook of a hanger when the hanger hook is placed over the hanging bar. The plurality of hanger detents can be configured to retain the hanger hook and/or to otherwise prevent translation of the hanger hook along a length of the hanging bar and provide a predetermined spacing of hangers.

Related U.S. Application Data

(60) Provisional application No. 63/630,672, filed on Feb. 26, 2024, provisional application No. 63/731,704, filed on Jun. 3, 2024.







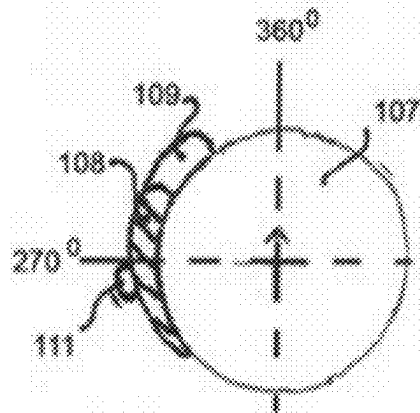


Fig 3

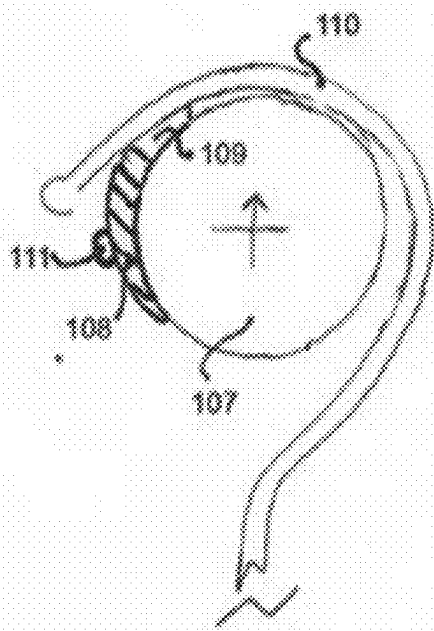


Fig 4

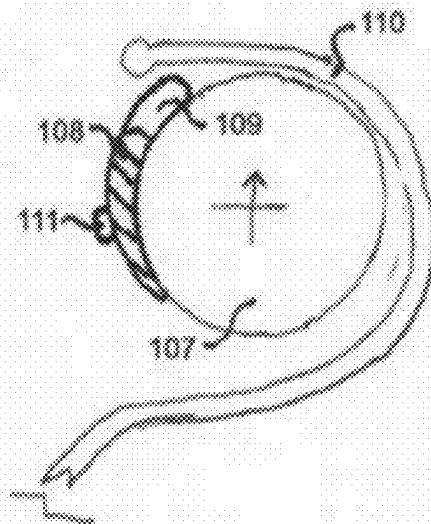


Fig 5

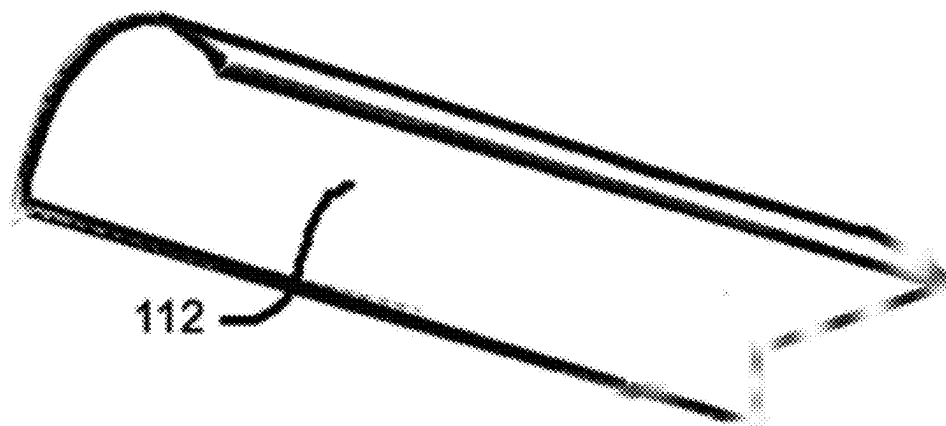


Fig. 6A

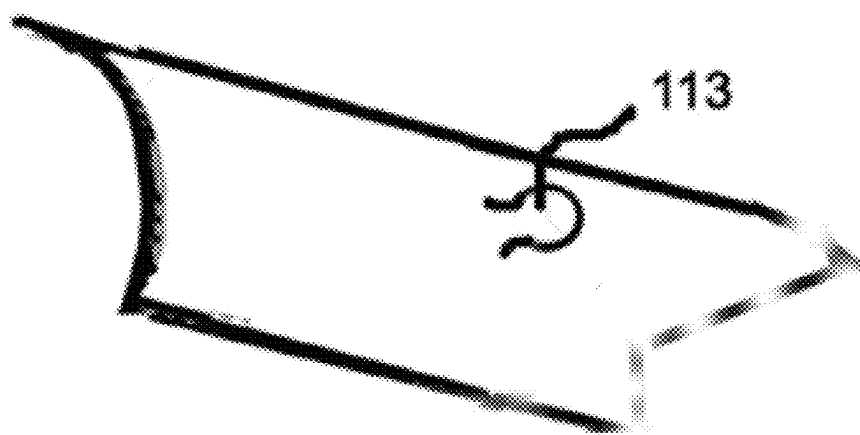


Fig. 6B

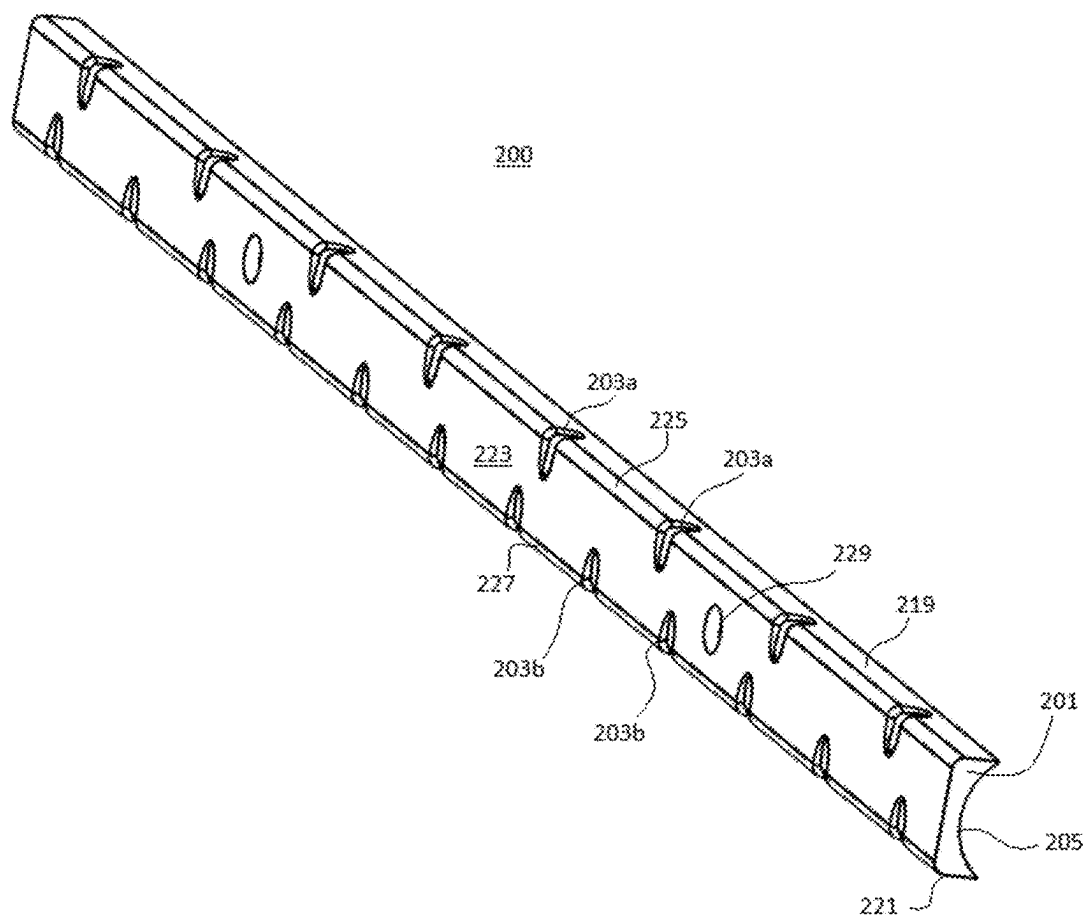


Fig. 7

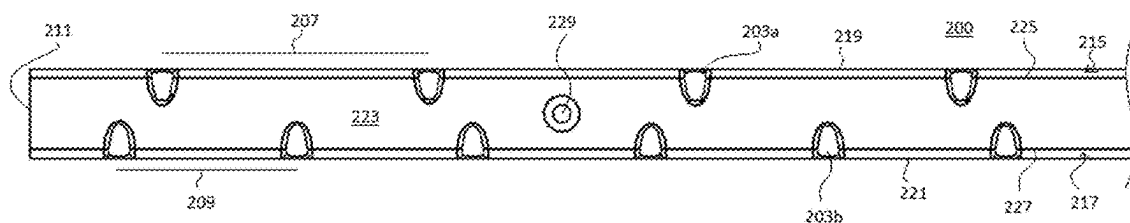


Fig. 8A

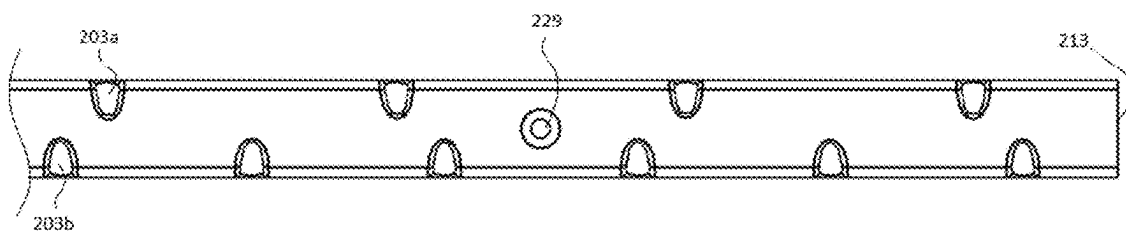


Fig. 8B

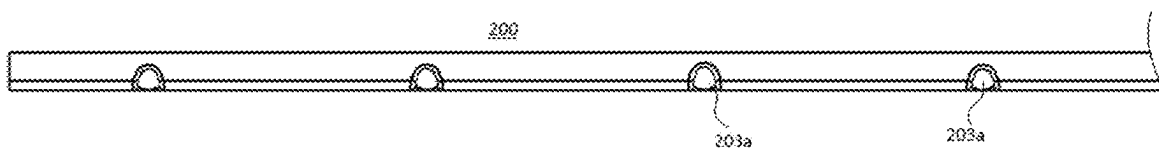


Fig. 9A

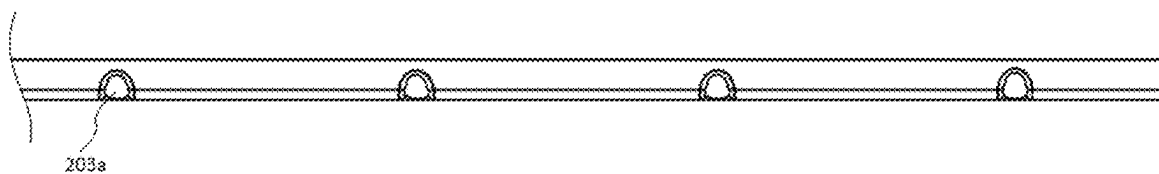


Fig. 9B

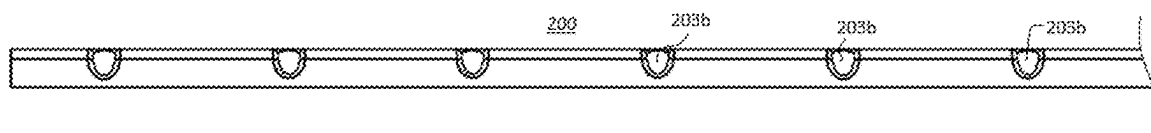


Fig. 10A

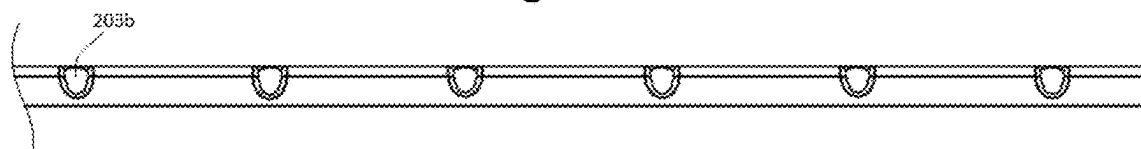


Fig. 10B

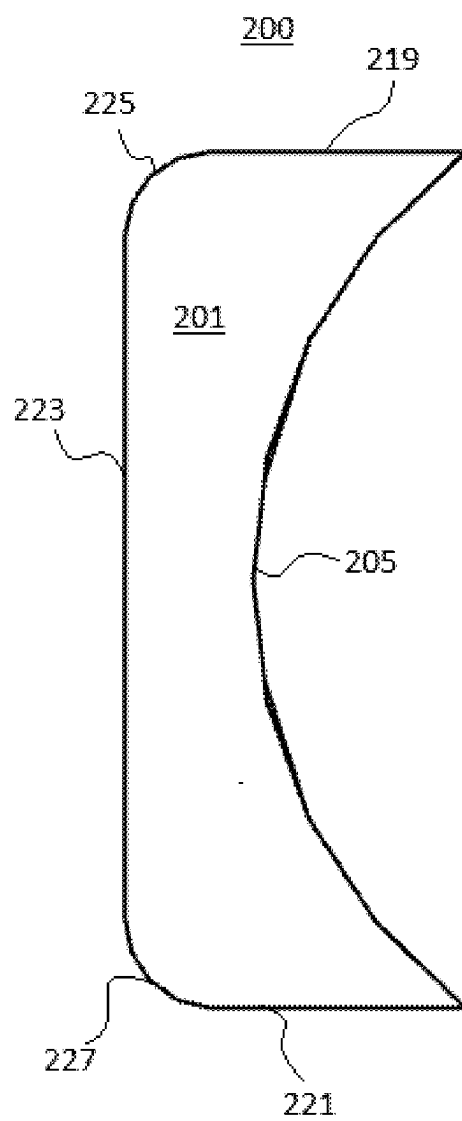


Fig. 11

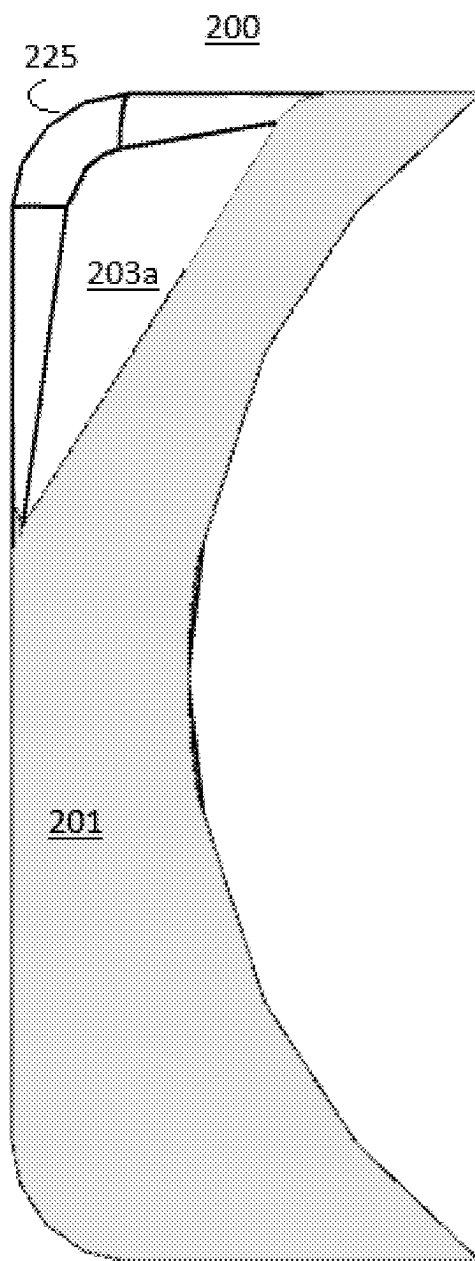


Fig. 12

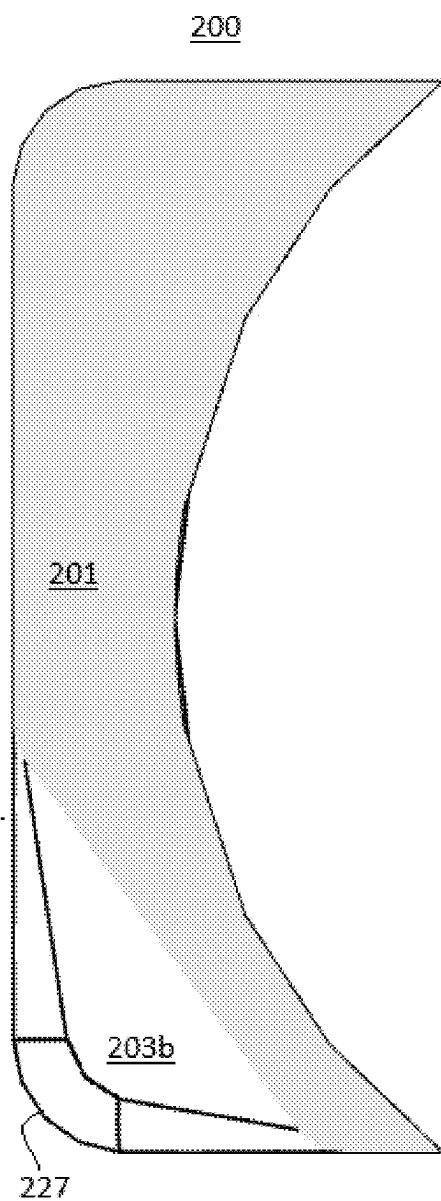


Fig. 13

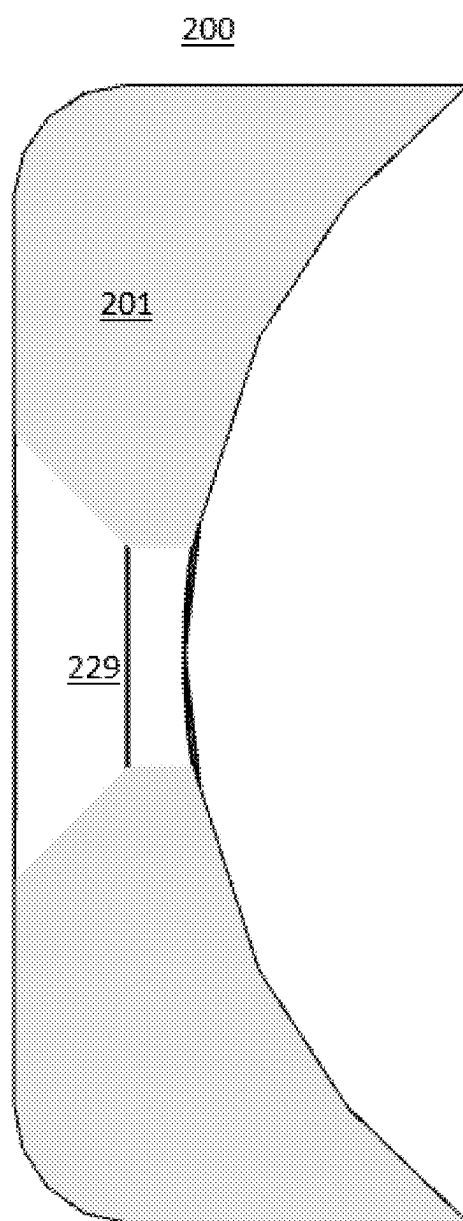


Fig. 14

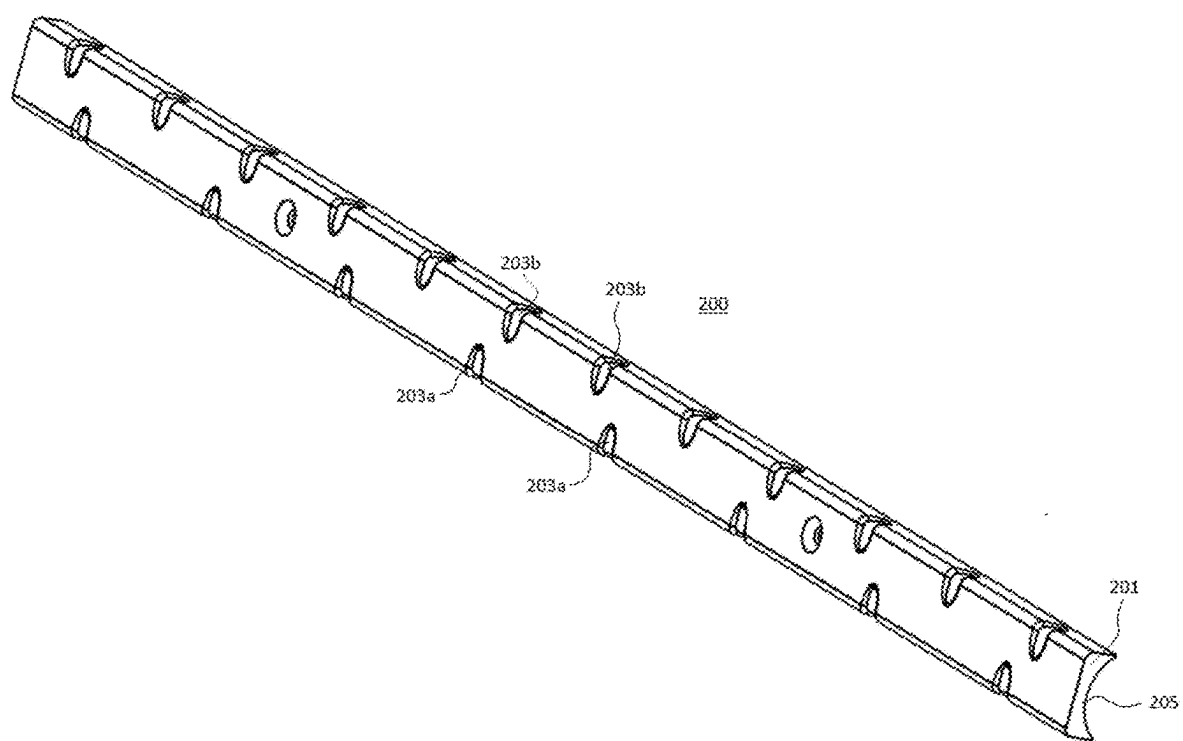


Fig. 15

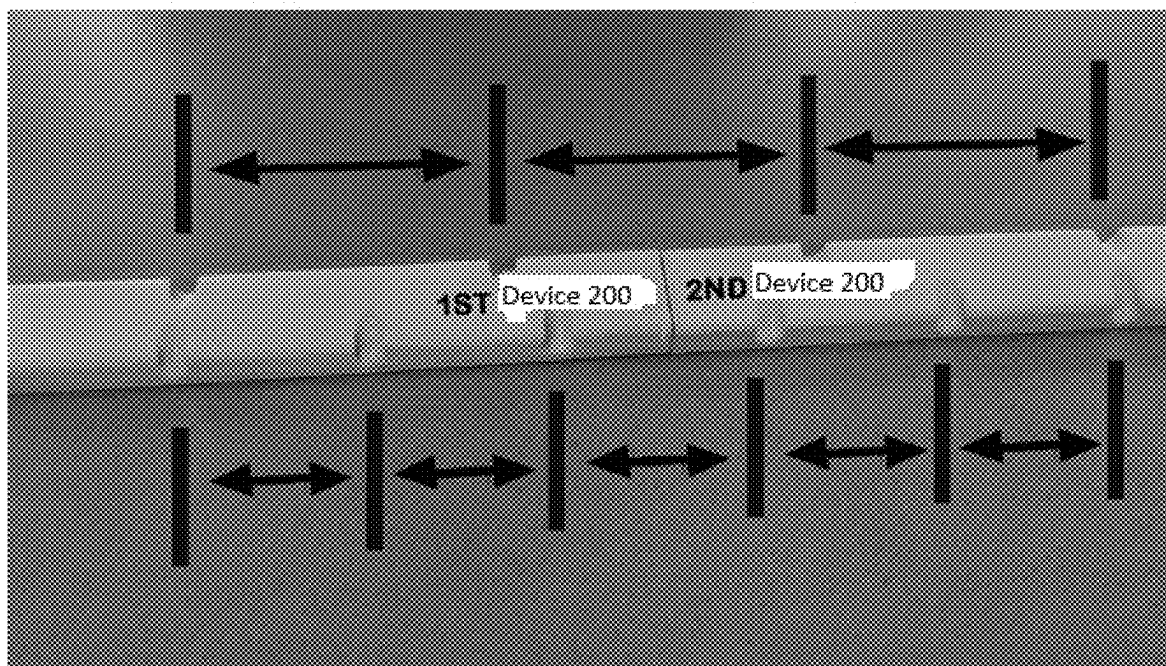


Fig. 16

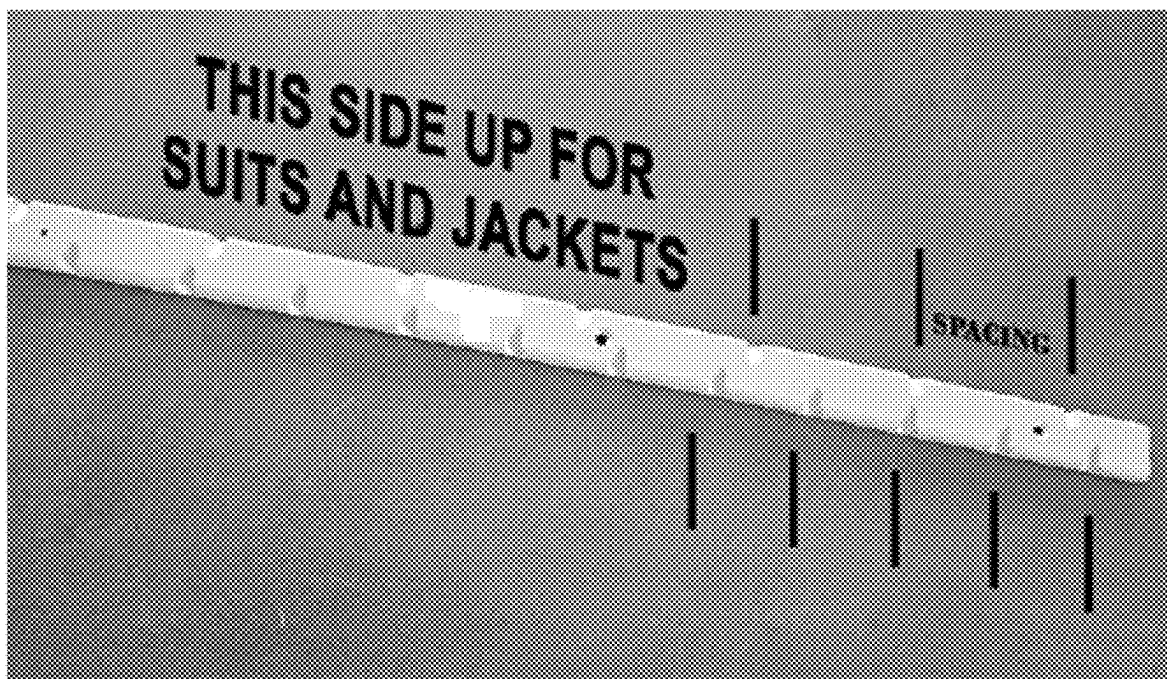


Fig. 17

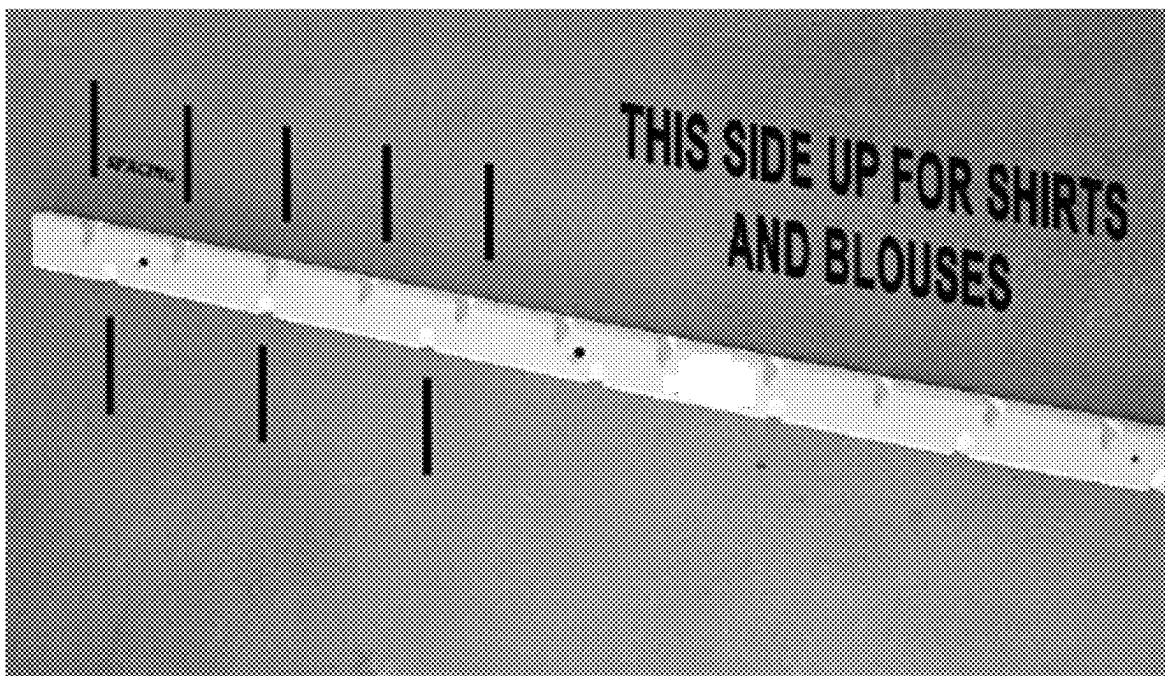


Fig. 18



Fig. 19



Fig. 20

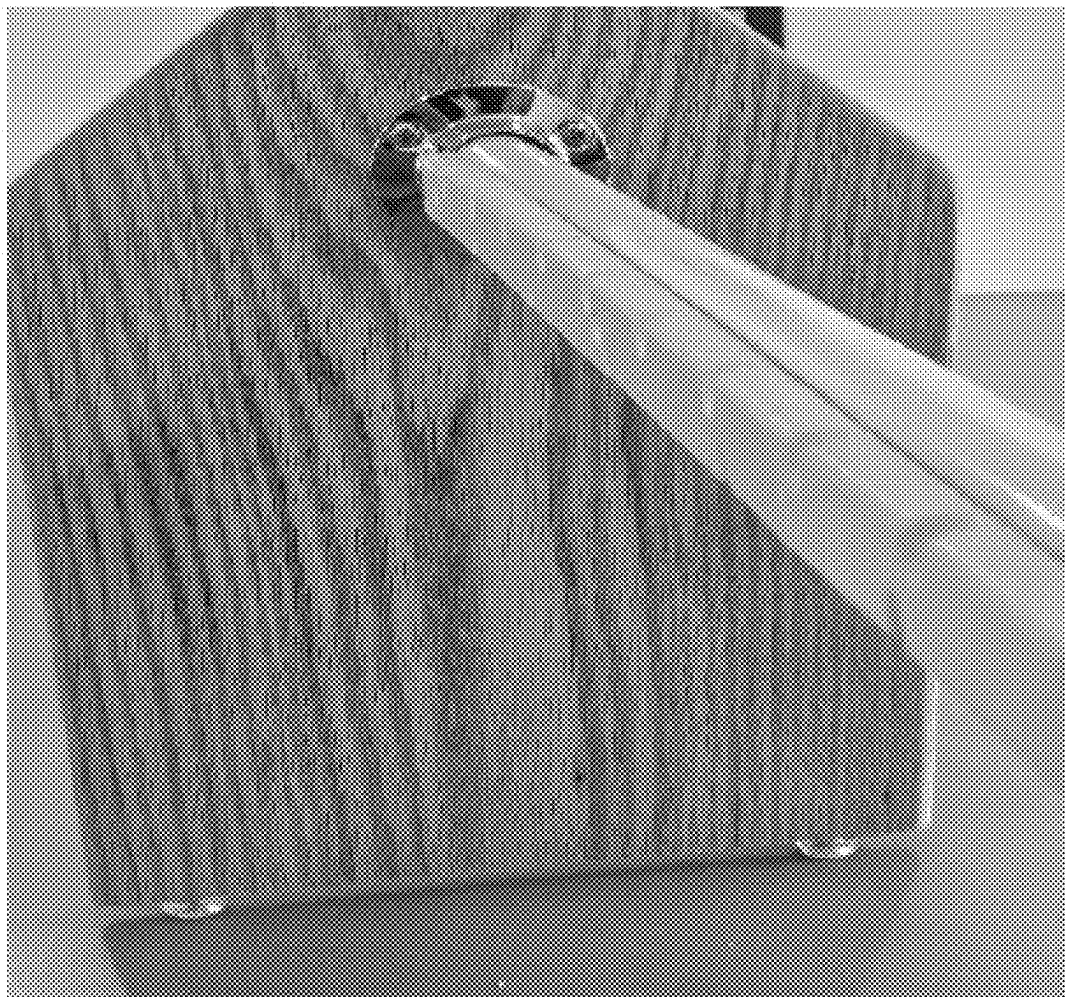


Fig. 21

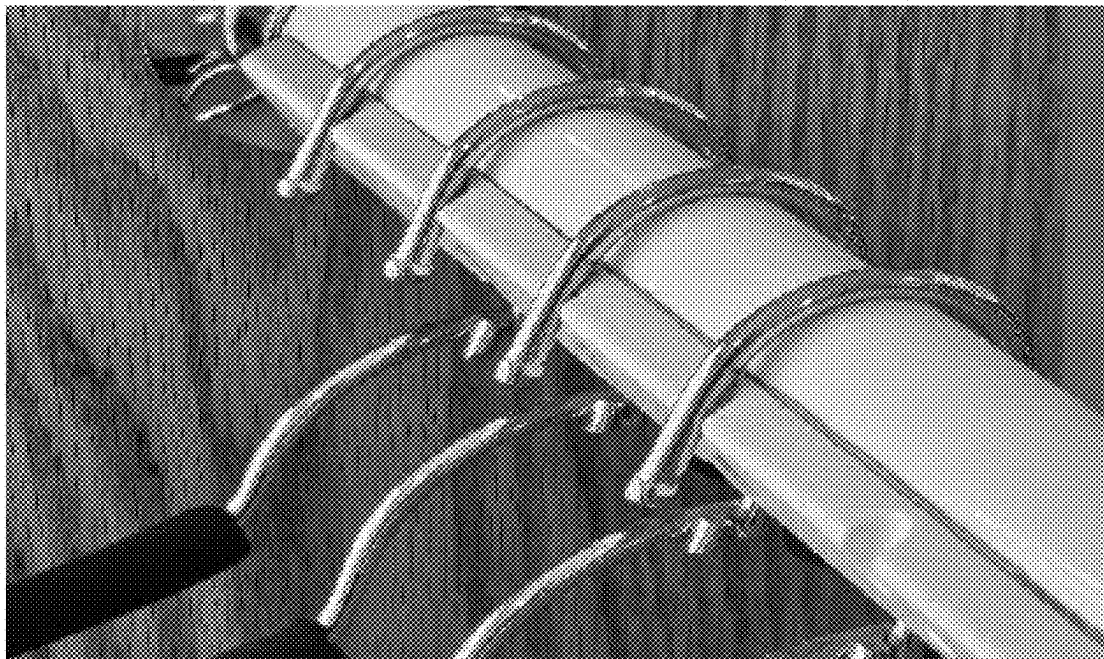


Fig. 22

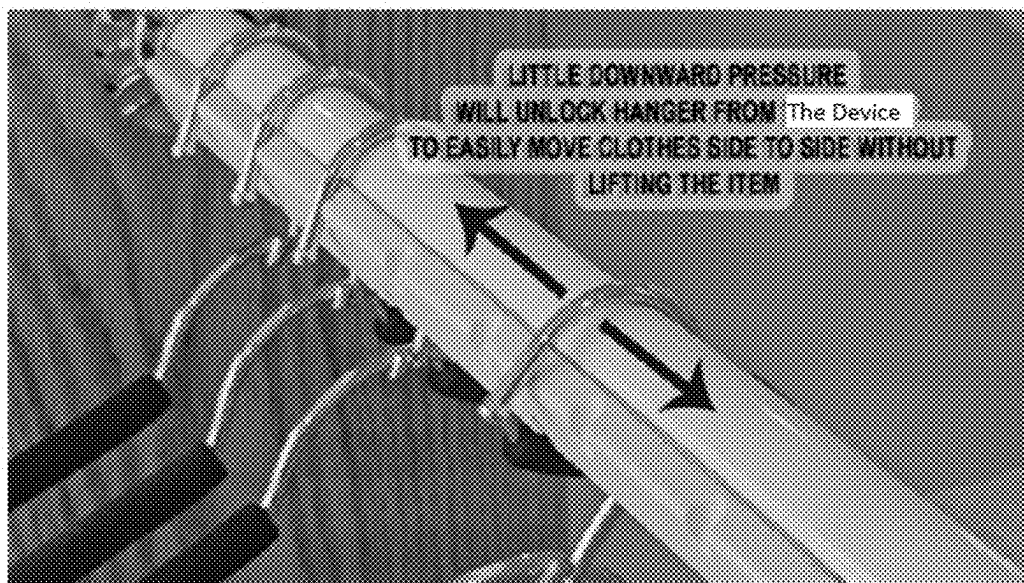


Fig. 23

HANGER DEVICES

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority to and the benefit of U.S. Provisional Application No. 63/630,672, filed Feb. 26, 2024, and U.S. Provisional Application No. 63/731,704, filed Jun. 3, 2024 the entire contents of each being herein incorporated by reference in their entirety.

FIELD

[0002] This disclosure relates to hanger devices.

BACKGROUND

[0003] Clothes hanging systems, e.g., in closets or elsewhere, typically consist of a hanging bar shaped to receive clothes-hangers for hanging clothes. Some traditional applications incorporate permanently fixed hanger separators or holes manufactured in the hanging bar that require the lifting of the garment and hanger completely from the hanging bar followed by the guess work of positioning the hanger to properly space garments when rearranging.

[0004] Such conventional methods and systems have generally been considered satisfactory for their intended purpose. However, there is still a need in the art for improvements. This disclosure provides a solution for this need.

SUMMARY

[0005] In accordance with at least one aspect of this disclosure, a device can include a body configured to attach to a hanging bar, and a plurality of hanger detents disposed in the body and configured to receive a portion of a hanger hook of a hanger when the hanger hook is placed over the hanging bar. The plurality of hanger detents can be configured to retain the hanger hook and/or to otherwise prevent translation of the hanger hook along a length of the hanging bar and provide a predetermined spacing of hangers.

[0006] The body can define a hanging bar opening configured to conform to a shape of the hanging bar. The hanging bar opening can be dimensioned to fit a circular hanging bar. The hanging bar opening can be dimensioned to fit a non-circular hanging bar. In certain embodiments, an adhesive disposed within the hanging bar opening.

[0007] The body can be configured to extend from the hanging bar. The body can be dimensioned to be at least partially hidden or totally hidden behind the hanging bar.

[0008] The plurality of detents can be disposed uniformly such that a predetermined spacing of hangers is uniform. The plurality of hanger detents can include two or more detents separated by a detent distance. In certain embodiments, a first detent of the plurality of hanger detents can be spaced about half of the detent distance from a first edge of the body, and a last detent of the plurality of detents can be spaced about half the distance from a second edge of the body such that placing one or more additional devices adjacent each other maintains the detent distance between detents of adjacent devices.

[0009] In certain embodiments, the plurality of hanger detents can include three or more detents. In certain embodiments, the plurality of hanger detents can include a first plurality of hanger detents disposed at a first perimeter portion of the body and a second plurality of hanger detents disposed at a second perimeter portion of the body. In certain

embodiments, the first plurality of hanger detents can be spaced a first detent distance from each other, and the second plurality of hanger detents can be spaced a second detent distance from each other, the first detent distance being different than the second detent distance.

[0010] In certain embodiments, the body can be invertible to allow a user to select use of either the first plurality of detents or the second plurality of detents. The body can be a symmetric shape about a lengthwise plane to allow uniform mounting to the hanging bar in either a first position or a second position upside-down relative to the first position.

[0011] In certain embodiments, the body can include a substantially flat first face, a substantially flat second face opposite the first face, a substantially flat third face that is substantially perpendicular to and extending between the first face and the second face, and a hanging bar opening defined by at least one hanging bar opening surface. In certain embodiments, the hanging bar opening can be positioned opposite the third face.

[0012] In certain embodiments, a first corner can be formed between the first face and the third face. In certain embodiments, a second corner can be formed between the second face and the third face. In certain embodiments, the first plurality of detents can be defined through the first corner between the first face and the third face, and the second plurality of detents can be defined through the second corner between the second face and the third face.

[0013] In certain embodiments, the at least one hanging bar opening surface can be a single smooth surface. In certain embodiments, one or more mounting holes can be defined through the third face and the at least one hanging bar opening surface can allow a fastener mount between the body and the hanging bar.

[0014] In accordance with at least one aspect of this disclosure, a hanging bar structure can include a hanging bar, and a device attached to or integrally forming part of the hanging bar. The device can be or include any embodiment of a device as disclosed herein, e.g., as described above. In certain embodiments, the body can extend away from a hanging bar centerline.

[0015] These and other features of the embodiments of the subject disclosure will become more readily apparent to those skilled in the art from the following detailed description taken in conjunction with the drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

[0016] So that those skilled in the art to which the subject disclosure appertains will readily understand how to make and use the devices and methods of the subject disclosure without undue experimentation, embodiments thereof will be described in detail herein below with reference to certain figures, wherein:

[0017] FIG. 1 is a perspective view of an embodiment of a device in accordance with this disclosure.

[0018] FIG. 2 shows the embodiment of FIG. 1 interacting and in position with an embodiment of a closet clothes hanging bar.

[0019] FIG. 3 depicts a cross-sectional view of the device where the cross section is taken through a centerline of an embodiment of a detent to reveal the detent position relative to the hanging bar.

[0020] FIG. 4 shows the embodiment of FIG. 3, depicting an embodiment of a clothes hanger hook interacting with a detent on the device locking/limiting the hanger from lateral movement.

[0021] FIG. 5 shows the embodiment of FIG. 3, depicting the clothes hanger with a slight rotation of the hook relative to the device and the hanging bar, showing the hook disengaged from the detent to allow lateral movement and repositioning of the hanger (e.g., by not being limited and being slid laterally along the bar).

[0022] FIGS. 6A and 6B depict two sides of an embodiment of a pressure sensitive strip showing the sticky side (FIG. 6A) and the opposite side (FIG. 6B) with one or more fastener protrusions that mate to one or more mounting holes that can snap-lock into mounting holes of the device for quick dismounting and mounting the device to the hanging bar.

[0023] FIG. 7 is a perspective view of an embodiment of a device in accordance with this disclosure.

[0024] FIGS. 8A and 8B show a plan view of the embodiment of FIG. 7.

[0025] FIGS. 9A and 9B show a first side elevation view of the embodiment of FIG. 7.

[0026] FIGS. 10A and 10B show a second side elevation view of the embodiment of FIG. 7, opposite the view of FIGS. 9A and 9B.

[0027] FIG. 11 is a lateral side elevation view of the embodiment of FIG. 7;

[0028] FIG. 12 is a cross-sectional view of the embodiment as shown in FIG. 7, taken through an embodiment of a detent of a first plurality of detents in accordance with this disclosure.

[0029] FIG. 13 is a cross-sectional view of the embodiment as shown in FIG. 7, taken through an embodiment of a detent of a second plurality of detent in accordance with this disclosure.

[0030] FIG. 14 is a cross-sectional view of the embodiment as shown in FIG. 7, taken through an embodiment of a mounting hole in accordance with this disclosure.

[0031] FIG. 15 is an upside down perspective view of the embodiment of FIG. 7.

[0032] FIG. 16 is a schematic diagram showing a plurality of devices abutting and maintaining spacing of detents.

[0033] FIG. 17 shows the embodiment of FIG. 16 in a first orientation for use of a first detent spacing.

[0034] FIG. 18 shows the embodiment of FIG. 16 in a second orientation, inverted from the first orientation of FIG. 17 for a different detent spacing.

[0035] FIG. 19 shows an embodiment of a peel and stick adhesive disposed in hanging bar opening and being peeled for attachment to a hanging bar.

[0036] FIG. 20 shows adhering the embodiment of FIG. 19 to the hanging bar.

[0037] FIG. 21 shows the embodiment of FIG. 20 attached to the hanging bar without any hangers thereon.

[0038] FIG. 22 shows the embodiment of FIG. 21 in use with a plurality of hangers engaged in detents of the device.

[0039] FIG. 23 shows the embodiment of FIG. 22 having a plurality of hangers thereon, one being rotated out of engagement with the device and free to slide along the hanging bar.

DETAILED DESCRIPTION

[0040] Reference will now be made to the drawings wherein like reference numerals identify similar structural features or aspects of the subject disclosure. For purposes of explanation and illustration, and not limitation, an illustrative view of an embodiment of a device in accordance with the disclosure is shown in FIG. 1 and is designated generally by reference character 99. Other views, embodiments, and/or aspects of this disclosure are illustrated in FIGS. 2-15.

[0041] Certain embodiments can include a clothes hanging system that retrofits to a hanging bar to arrange clothes by indexing hangers with indentations to resist hanger motion from side-to-side relative to the hanging bar and can change index spacing by rotating the system one hundred and eighty degrees relative to the hanging bar, and can be disengaged with a downward motion on one end of hanger to rotate the hanger hook to a position that dislodges the hanger from its detent allowing the hanger to move freely across the hanging bar without lifting the hanger from the bar.

[0042] In certain embodiments, a clothes hanging system can include a hanging bar made from a round bar of metal, wood or any other suitable material, and a device (e.g., a spacing indexer) made of metal, plastic, or any other suitable material. In certain embodiments, the device can be in the form of a linear bar, e.g., with at least four sides. In certain embodiments, the device's height can be pre-determined to be no greater than the diameter of the hanging bar, and its length no longer than the hanging bar. In certain embodiments, one of the device's sides is concaved to mate lengthwise with the circumference of the hanging bar, and the thickness of one or two sides of the device can be predetermine to accept detents (e.g., notches) that can receive the thickness of typical clothes-hangers and then have multiple detents suitably spaced the full length of the device on one or both edges. In certain embodiments, the device's upper and lower edge is at least sized to receive detents to fit a portion of the clothes hangers' hook plus mounting screw holes. In certain embodiments, the upper edge of the device with the detents is mounted in close proximity to highest elevation point of a hanging bar and within the nine and twelve o'clock position or in other terms between two hundred and seventy degrees to three hundred sixty degrees of the circumference of the hanging bar, and one edge of the width of the device can be tapered convexly to maintain the integrity of the device while keeping a low profile to avoid interfering with hangers.

[0043] In certain embodiments, the device is strategically mounted to the hanging bar so that when a clothes hanger is at rest on the hanging bar a portion of the hanger's hook is engaged with a detent restricting lateral movement. In certain embodiments, a slight downward push on the end of the hanger causes the hanger's hook to rotate on the hanging bar freeing the hook from the detent to allow lateral movement of the hanger without lifting the weight of the hanger and garment. Releasing the hanger allows the hook to rotate back to its original center of gravity, allowing reengagement of the hanger to the nearest detent with slight sideways movement, thus properly spacing garments to keep clothes fresh and free of wrinkles.

[0044] In certain embodiments, the device can be fitted with pressure sensitive glues affixed to the concave surface as an alternative to screw mounting to hanging bar. In certain embodiments, the device can be mounted to a mounting strip

consisting of a tape-on component with fasteners that are positioned to mate with mounting holes on the device, allowing detachment and reattachment after a 180 degrees rotation of the device to another detent scheme.

[0045] In certain embodiments, the device can be mounted with screws to be able to change the device to another detent scheme, when different arrangement is needed e.g., for jackets and suits vs shirts. In certain embodiments, the device can have strategic positioning of the detents to seamlessly add additional indexers for longer hanging lengths while maintaining detent interval spacing.

[0046] Referring to the embodiment of FIGS. 1-6B generally, a device **99** can include a concave side **100** that can mate to the circumference of the hanging bar **107**, e.g., as shown in FIGS. 2, 3, 4 and 5. The thickness **101** of the device can give debt for the detents **102**. One or more mounting holes **103** can include drilled, or partially drilled, screw holes for optional mounting that allow for the device's removal and replacement. A predetermined spacing **104** between detents can dictates how close garments can be hung next to each other. Half spacing **105** at the ends of the device **99** can allow for two devices **99** to be mounted together lengthwise without interrupting a continuous spacing of the garments. A convex shaped side **106** of the device **99** can allow for a lower profile of the device **99** so as not to interfere with hangers while maintaining the integrity of the device for mounting purposes.

[0047] Referring to FIG. 2, an embodiment of a hanging bar **109** is shown positioned in relation with the concaved side **100** of the device **99**. Referring to FIG. 3, a cross-section material **108** is shown in relation to detent **109** and in relationship with hanging bar **107**. The detent **109** can be positioned between about 270 degrees and 360 degrees of the circumference of the hanging bar **107**. In certain embodiments, the device **99** can include any suitable optional mounting fasteners **111** (e.g., screws).

[0048] Referring to FIG. 4, an embodiment of a clothes hanger reference **110** is shown at rest hanging on the hanging bar **107** with the hook of the hanger **110** engaged in a detent **109** locking and/or limiting the hanger **110** from lateral movement. Referring to FIG. 5, the hook of the hanger **110** can be rotated about the hanging bar **107** and thereby disengaging from the detent **109** allowing the hanger **110** to move side-to-side and relocate to a different detent **109** on the device **99**, for example. Referring to FIGS. 6A and 6B, a pressure sensitive strip **112** can include a sticky adhesive side (FIG. 6A) of strip and protrusions **113** that can mate and fasten to mounting holes of the device **99** for dismounting and remounting of the device **99**.

[0049] Embodiments can allow a user, with little effort, to use the length of a hanger to leverage the hanger at fulcrum point where the hook pivots on the hanging bar to cause rotation of the hanger's hook to disengage the detent and reposition the hanger by sliding it to another detent and reengage a detent by letting the hanger rest to its natural center of balance position.

[0050] Embodiments can include a clothes organizer that comprises a device that is precisely fixed lengthwise on a typical spherical clothes hanging bar with a linear concaved side that interact with the circumference of the hanging bar, and has predetermined detents positioned lengthwise to receive typical clothes hangers and is positioned relative to the hanging bar to allow disengagement of detents without separating hanger from the hanging bar. Certain embodi-

ments can include mounting holes for mounting screws. Certain embodiments can include mounting pressure sensitive glues or tapes. Certain embodiments can include a convex tapered side for a lower profile to avoid unnecessary interference with hangers. Certain embodiments can include a predetermined spacing of detents and distancing to allow linking two or more devices without interrupting arrangements. Certain embodiments can include a pressure sensitive mounting strip with fasteners to mate and lock with said mounting holes

[0051] Certain embodiments can include a clothes hanging system that retrofits to a hanging bar to arrange clothes by indexing hangers with indentations to resist hanger motion from side-to-side relative to the bar and can be disengaged with a downward motion on one end of hanger to rotate the hanger hook to a position that dislodges the hanger from its detent allowing the hanger to move freely across the hanging bar without lifting the hanger from the bar.

[0052] Certain embodiments relate to a clothes arrangement system used with clothes hangers and hanging bar. Certain embodiments can include a clothes hanging system can include a hanging bar made from a round bar of metal, wood or any other suitable material, a spacing indexer comprising a device, made of metal, plastic, or like materials, in the form of a linear bar with at least three sides, whereas the device's width is predetermined to be no greater than the diameter of the hanging bar, and its length no longer than the hanging bar and whereas one of the device's sides is concaved to mate lengthwise with the circumference of the hanging bar, and where the thickness of one side of the device is predetermine to accept detents that can receive the thickness of typical clothes-hangers and then have multiple detents formed with knock-out plugs, for the user to customize hanger separation, and that are evenly spaced the full length of the device. In certain embodiments, the device is at least sized widthwise to receive the detent, that are sized to fit a portion of a clothes hanger hook, plus mounting screw holes. In certain embodiments, the thickest side of the device with the detents is mounted in close proximity to a highest elevation point of a hanging bar and within the nine and twelve o'clock position or in other terms between two hundred and seventy degrees to 360 degrees of the diameter of the hanging bar. In certain embodiments, another edge of the width of the device can be tapered convexly to maintain the integrity of the device while keeping a low profile to avoid interfering with hangers.

[0053] In certain embodiments, the device can be strategically mounted to the hanging bar so that when a clothes hanger is at rest on the hanging bar a portion of the hanger's hook is engaged with a detent, that has had the knock-out removed, restricting lateral movement. In certain embodiments, a slight downward push on the end of the hanger causes the hanger's hook to rotate on the hanging bar freeing the hook from the detent to allow lateral movement of the hanger without lifting the weight of the hanger and garment. In certain embodiments, releasing the hanger allows the hook to rotate back to its original center of gravity, allowing reengagement of the hanger to the nearest opened detent avoiding detents with knock-outs with sideways movement, thus properly spacing garments at desired intervals to keep clothes fresh and free of wrinkles.

[0054] In certain embodiments, the device can be fitted with pressure sensitive glues affixed to the concave surface

as an alternative to screw mounting to hanging bar. In certain embodiments, the device can be mounted with screws to be able to change the device to another one with different detent intervals, when different arrangement is needed e.g., for jackets and suits vs shirts. In certain embodiments, the device can have strategic positioning of the detents to seamlessly add additional indexers for longer hanging lengths while maintaining detent interval spacing.

[0055] In certain embodiments, the device can include a detent enabled by the removal of its knock-out plug and the rest disabled with their knock-out plugs intact. In certain embodiments, all detents enable interacting and positioning with a typical closet clothes hanging bar.

[0056] In certain embodiments, a concave side of the device can mate to the circumference of the hanging bar, and the thickness of the device can give depth for the detents as illustrated as an opened detent. In certain embodiments, a knocked out plug can enable a detent while the rest of the plugs can be still intact with their respective detents. Certain embodiments can include holes that are drilled, or partially drilled, screw holes for optional mounting that allow for the device's removal and replacement. Certain embodiments can include a predetermine spacing between detents that dictates how close garments can be hung next to each other. Certain embodiments can include a half spacing at the ends of the device to allows for two devices to be mounted together lengthwise without interrupting a continuous spacing of the garments. Certain embodiments can include a convex shaped side of the device which allows for a lower profile of the device so as not to interfere with hangers while maintaining the integrity of the device for mounting purposes.

[0057] Certain embodiments can include a clothes organizer comprising a device that is precisely fixed lengthwise on a typical spherical clothes hanging bar with a linear concaved side that interact with the circumference of the hanging bar, and has predetermined detents positioned length-wise to receive typical clothes hangers and is positioned relative to the hanging bar to allow disengagement of detents without separating hanger from the hanging bar. Certain embodiments can include controlled spacing using knock out plugs to enable detents. Certain embodiments can include mounting holes. Certain embodiments can include mounting pressure sensitive glues or tapes. Certain embodiments can include a convex tapered side. Certain embodiments can include predetermined spacing of detents linking two or more devices without interrupting arrangements.

[0058] With reference to the embodiment shown in FIGS. 7-15, in accordance with at least one aspect of this disclosure, a device 200 can include a body 201 configured to attach to a hanging bar (e.g., hanging bar 107 as shown in FIGS. 2-5). The device 200 can include a plurality of hanger detents 203a, 203b disposed in the body 201 and configured to receive a portion of a hanger hook of a hanger (e.g., hanger 110 as shown in FIGS. 3-5) when the hanger hook is placed over the hanging bar. The plurality of hanger detents 203a, 203b can be configured to retain the hanger hook and/or to otherwise prevent translation (e.g., sliding) of the hanger hook along a length (e.g., in a lengthwise direction) of the hanging bar and provide a predetermined spacing of hangers.

[0059] The body 201 can define a hanging bar opening 205 configured to conform to a shape of the hanging bar

(e.g., a partially circular shape to receive a round bar), e.g., to provide a flush contact. Any other suitable shape is contemplated herein.

[0060] The hanging bar opening 205 can be dimensioned to fit a circular hanging bar. In certain embodiments, the hanging bar opening 205 can be dimensioned to fit a non-circular hanging bar (e.g., a square bar, an oval bar, etc.). In certain embodiments, an adhesive (e.g., tape, glue, etc.) can be disposed within the hanging bar opening 205 (e.g., to allow adhering to the hanging bar). In certain embodiments, the hanging bar opening 205 can include any one or more of any other suitable fastener or device to allow temporary connection or permanent connection to a hanger bar.

[0061] The body 201 can be configured to extend from the hanging bar (e.g., radially outwardly from a rear portion of a round bar). In certain embodiments, the body 201 can be dimensioned to be at least partially hidden or totally hidden behind the hanging bar (e.g., at least when looking at the hanger bar at eye level, and/or within about 45 degrees from eye level).

[0062] As shown, the plurality of detents 203a, 203b can be disposed uniformly such that a predetermined spacing of hangers is uniform. In certain embodiments, the plurality of hanger detents 203a, 203b can include two or more detents (e.g., detents 203a or detents 203b) separated by a detent distance (e.g., a first detent distance 207 for detents 203a, and a second detent distance 209 for detents 203b). In certain embodiments, a first detent 203a, 203b (e.g., positioned first from a first edge 211) of the plurality of hanger detents 203a, 203b can be spaced about half of the detent distance 207, 209 from a first edge 211 of the body 201, and a last detent (e.g., a first detent 203a, 203b from a second edge 213) of the plurality of detents 203a, 203b can be spaced about half the distance from a second edge 213 of the body 201 such that placing one or more additional devices 200 adjacent each other maintains the detent distance 207, 209 between detents 203a, 203b of adjacent devices 200.

[0063] In certain embodiments, the plurality of hanger detents 203a, 203b can include three or more detents 203a, 203b. In certain embodiments, the plurality of hanger detents 203a, 203b can include a first plurality of hanger detents 203a disposed at a first perimeter portion 215 of the body 201 (e.g., along a first edge line) and a second plurality of hanger detents 203b disposed at a second perimeter portion 217 of the body 201. In certain embodiments, the first plurality of hanger detents 203a can be spaced a first detent distance 207 from each other, and the second plurality of hanger detents 203b can be spaced a second detent distance 209 from each other. The first detent distance 207 can be different (e.g., larger) than the second detent distance 209.

[0064] In certain embodiments, the body 201 can be invertible to allow a user to select use of either the first plurality of detents 203a or the second plurality of detents 203b. As shown, the body 201 can be a symmetric shape about a lengthwise plane to allow uniform mounting to the hanging bar in either a first position (e.g., as shown in FIG. 8) or a second position upside-down relative to the first position (e.g., as shown in FIG. 15).

[0065] In certain embodiments, the body 201 can include a substantially flat first face 219, a substantially flat second face 221 opposite the first face, a substantially flat third face 223 that is substantially perpendicular to and extending

between the first face **219** and the second face **221**, and the hanging bar opening **205** defined by at least one hanging bar opening surface. In certain embodiments, the hanging bar opening **205** can be positioned opposite the third face **223**.

[0066] In certain embodiments, a first corner **225** (e.g., a beveled or chamfered edge, a straight edge, a rounded edge, etc.) can be formed between the first face **219** and the third face **223**. In certain embodiments, a second corner **227** (e.g., a beveled or chamfered edge, a straight edge, a rounded edge etc.) can be formed between the second face **221** and the third face **223**. In certain embodiments, the first plurality of detents **203a** can be defined through the first corner **225** between the first face **219** and the third face **223**, and the second plurality of detents **203b** can be defined through the second corner **227** between the second face **221** and the third face **223**.

[0067] In certain embodiments, the at least one hanging bar opening **205** can be defined by a single smooth surface (e.g., a curved surface as shown, or any other suitable shape, conforming or otherwise). In certain embodiments, one or more mounting holes **229** can be defined through the third face **223** and the at least one hanging bar opening surface (that defines hanging bar opening **205**) can allow a fastener mount (e.g., screws or other fasteners) between the body **201** and the hanging bar. Any other suitable mechanism to allow detachable mounting is contemplated herein (e.g., a clip system, a multipiece system, etc.).

[0068] In accordance with at least one aspect of this disclosure, a hanging bar structure can include a hanging bar (e.g., hanging bar **107** or any other suitable hanging bar of any suitable shape), and a device (e.g., device **200**) attached to or integrally forming part of (e.g., formed with) the hanging bar. The device can be or include any embodiment of a device as disclosed herein, e.g., a device **200** as described above. In certain embodiments, the body (e.g., body **201**) can extend away from a hanging bar centerline.

[0069] FIG. **16** is a schematic diagram showing a plurality of devices abutting and maintaining spacing of detents. FIG. **17** shows the embodiment of FIG. **16** in a first orientation for use of a first detent spacing. FIG. **18** shows the embodiment of FIG. **16** in a second orientation, inverted from the first orientation of FIG. **17** for a different detent spacing. FIG. **19** shows an embodiment of a peel and stick adhesive disposed in hanging bar opening and being peeled for attachment to a hanging bar. FIG. **20** shows adhering the embodiment of FIG. **19** to the hanging bar. FIG. **21** shows the embodiment of FIG. **20** attached to the hanging bar without any hangers thereon. FIG. **22** shows the embodiment of FIG. **21** in use with a plurality of hangers engaged in detents of the device. FIG. **23** shows the embodiment of FIG. **22** having a plurality of hangers thereon, one being rotated out of engagement with the device and free to slide along the hanging bar.

[0070] Embodiments can enable connecting additional devices without losing the spacing of garments as long as edges are abutted or otherwise connected (which any such suitable a connection can be included in the body). Certain embodiments of a device can be cut to a desired size. In certain embodiments, the spacing of detents can be changed by mounting the device flipped 180 degrees. Certain embodiments can include a peel and stick mounting system, and can hang out of sight behind a clothes hanging bar/rod. With a slight downward pressure on hanger to rotate the

hanger hook slightly, the device be configured to disengage the hanger hook to slide clothes back and forth along the hanging bar.

[0071] Those having ordinary skill in the art understand that any numerical values disclosed herein can be exact values or can be values within a range. Further, any terms of approximation (e.g., “about”, “approximately”, “around”) used in this disclosure can mean the exact stated value and the stated value within a range. For example, in certain embodiments, for the stated values disclosed herein, the range is within (plus or minus) 20% of the stated value. In certain embodiments of the stated values disclosed herein, the range is within (plus or minus) 10% of the stated value. In certain embodiments, for the stated values disclosed herein, the range is within (plus or minus) 5% of the stated value. In certain embodiments, for the stated values disclosed herein, the range is within (plus or minus) 2% of the stated value. In certain embodiments, for the stated values disclosed herein, the range is within (plus or minus) 1% of the stated value. In certain embodiments, for the stated values disclosed herein, the range is within (plus or minus) 0.5% of the stated value. In certain embodiments, for the stated values disclosed herein, the range is within (plus or minus) 0.25% of the stated value. In certain embodiments, for the stated values disclosed herein, the range is within (plus or minus) 0.1% of the stated value. In certain embodiments, for the stated values disclosed herein, the range is within (plus or minus) 0.05% of the stated value. In certain embodiments, for the stated values disclosed herein, the range is within (plus or minus) 0.01% of the stated value. In certain embodiments, for the stated values disclosed herein, the range is within (plus or minus) 0.001% of the stated value. In certain embodiments, for the stated values disclosed herein, the range is within any other suitable percentage or number as appreciated by those having ordinary skill in the art (e.g., for known tolerance limits or error ranges). In certain embodiments, any combination of the above ranges for any combination of stated values are contemplated herein.

[0072] The articles “a”, “an”, and “the” as used herein and in the appended claims are used herein to refer to one or more than one (i.e., to at least one) of the grammatical object of the article unless the context clearly indicates otherwise. By way of example, “an element” means one element or more than one element.

[0073] The phrase “and/or,” as used herein in the specification and in the claims, should be understood to mean “either or both” of the elements so conjoined, i.e., elements that are conjunctively present in some cases and disjunctively present in other cases. Multiple elements listed with “and/or” should be construed in the same fashion, i.e., “one or more” of the elements so conjoined. Other elements may optionally be present other than the elements specifically identified by the “and/or” clause, whether related or unrelated to those elements specifically identified. Thus, as a non-limiting example, a reference to “A and/or B”, when used in conjunction with open-ended language such as “comprising” can refer, in one embodiment, to A only (optionally including elements other than B); in another embodiment, to B only (optionally including elements other than A); in yet another embodiment, to both A and B (optionally including other elements); etc.

[0074] As used herein in the specification and in the claims, “or” should be understood to have the same meaning

as “and/or” as defined above. For example, when separating items in a list, “or” or “and/or” shall be interpreted as being inclusive, i.e., the inclusion of at least one, but also including more than one, of a number or list of elements, and, optionally, additional unlisted items. Only terms clearly indicated to the contrary, such as “only one of” or “exactly one of,” or, when used in the claims, “consisting of,” will refer to the inclusion of exactly one element of a number or list of elements. In general, the term “or” as used herein shall only be interpreted as indicating exclusive alternatives (i.e., “one or the other but not both”) when preceded by terms of exclusivity, such as “either,” “one of,” “only one of,” or “exactly one of.”

[0075] Any suitable combination(s) of any disclosed embodiments and/or any suitable portion(s) thereof are contemplated herein as appreciated by those having ordinary skill in the art in view of this disclosure.

[0076] The embodiments of this disclosure, as described above and shown in the drawings, provide for improvement in the art to which they pertain. While the subject disclosure includes reference to certain embodiments, those skilled in the art will readily appreciate that changes and/or modifications may be made thereto without departing from the spirit and scope of the subject disclosure.

What is claimed is:

1. A device, comprising:
 - a body configured to attach to a hanging bar;
 - a plurality of hanger detents disposed in the body and configured to receive a portion of a hanger hook of a hanger when the hanger hook is placed over the hanging bar, wherein the plurality of hanger detents are configured to retain the hanger hook and/or to otherwise prevent translation of the hanger hook along a length of the hanging bar and provide a predetermined spacing of hangers.
2. The device of claim 1, wherein the body defines a hanging bar opening configured to conform to a shape of the hanging bar.
3. The device of claim 2, wherein the hanging bar opening is dimensioned to fit a circular hanging bar.
4. The device of claim 2, wherein the hanging bar opening is dimensioned to fit a non-circular hanging bar.
5. The device of claim 2, further comprising an adhesive disposed within the hanging bar opening.
6. The device of claim 1, wherein the body is configured to extend from the hanging bar.
7. The device of claim 6, wherein the body is dimensioned to be at least partially hidden or totally hidden behind the hanging bar.
8. The device of claim 1, wherein the plurality of detents are disposed uniformly such that a predetermined spacing of hangers is uniform.
9. The device of claim 8, wherein the plurality of hanger detents includes two or more detents separated by a detent distance, a first detent of the plurality of hanger detents being spaced about half of the detent distance from a first edge of the body, and a last detent of the plurality of detents being spaced about half the distance from a second edge of

the body such that placing one or more additional devices adjacent each other maintains the detent distance between detents of adjacent devices.

10. The device of claim 1, wherein the plurality of hanger detents includes three or more detents.

11. The device of claim 1, wherein the plurality of hanger detents include a first plurality of hanger detents disposed at a first perimeter portion of the body and a second plurality of hanger detents disposed at a second perimeter portion of the body.

12. The device of claim 11, wherein the first plurality of hanger detents are spaced a first detent distance from each other, and the second plurality of hanger detents are spaced a second detent distance from each other, the first detent distance being different than the second detent distance.

13. The device of claim 12, wherein the body is invertible to allow a user to select use of either the first plurality of detents or the second plurality of detents.

14. The device of claim 13, wherein the body has a symmetric shape about a lengthwise plane to allow uniform mounting to the hanging bar in either a first position or a second position upside-down relative to the first position.

15. The device of claim 14, wherein the body comprises a substantially flat first face, a substantially flat second face opposite the first face, a substantially flat third face that is substantially perpendicular to and extending between the first face and the second face, and a hanging bar opening defined by at least one hanging bar opening surface, the hanging bar opening positioned opposite the third face.

16. The device of claim 15, wherein a first corner is formed between the first face and the third face, wherein a second corner is formed between the second face and the third face, wherein the first plurality of detents are defined through the first corner between the first face and the third face, and the second plurality of detents are defined through the second corner between the second face and the third face.

17. The device of claim 16, wherein the at least one hanging bar opening surface is a single smooth surface.

18. The device of claim 16, further comprising one or more mounting holes defined through the third face and the at least one hanging bar opening surface to allow a fastener mount between the body and the hanging bar.

19. A hanging bar structure, comprising:

- a hanging bar; and
- a device attached to or integrally forming part of the hanging bar, the device comprising:
 - a body; and
 - a plurality of hanger detents disposed in the body and configured to receive a portion of a hanger hook of a hanger when the hanger hook is placed over the hanging bar, wherein the plurality of hanger detents are configured to retain the hanger hook and/or to otherwise prevent translation of the hanger hook along a length of the hanging bar and provide a predetermined spacing of hangers.

20. The hanging bar structure of claim 19, wherein the body extends away from a hanging bar centerline.

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